

1. SCHUBERT CALCULUS ON THE GRASSMANNIAN

In this appendix we develop the Schubert calculus on the complex Grassmannian, beginning with the Plucker embedding and ending with the interpretation of Chern classes as Schubert cycles. The primary references are Fulton [Ful97] and Eisenbud–Harris [EH16].

1.1. The Plücker embedding. Throughout this section, let F be an algebraically closed field, let V be an n -dimensional F -vector space, and let $\text{Gr}(k, V) = \text{Gr}(k, n)$ denote the Grassmannian of k -dimensional subspaces of V . Our goal is to show $\text{Gr}(k, V)$ is a projective variety via the Plucker embedding.

Suppose that $W \subset V$ is a k -dimensional linear subspace spanned by $v_1, \dots, v_k$. We form the wedge

$$v_1 \wedge \cdots \wedge v_k \in \bigwedge^k(V),$$

which is an element in the k -th exterior power of V . This is a non-zero element because the v_i are linearly independent. Note that if you choose a different basis for V , the wedge would change only by a nonzero scalar factor, which is given by the determinant of the change of basis matrix. Hence,

$$[v_1 \wedge \cdots \wedge v_k] \in \mathbb{P} \left(\bigwedge^k(V) \right).$$

This gives us the well defined map:

Definition 1.1. The *Plucker map* is the map

$$\begin{aligned} \psi : \text{Gr}(k, n) &\longrightarrow \mathbb{P} \left(\bigwedge^k V \right) \\ W = \text{span}(v_1, \dots, v_k) &\longmapsto [v_1 \wedge \cdots \wedge v_k]. \end{aligned}$$

For any $[\omega] = \psi(W)$, we can uniquely reconstruct W from the wedge product of its basis vectors. In particular, it is the space of $v \in V$ such that $v \wedge \omega = 0 \in \bigwedge^{k+1} V$. Thus, it is also an inclusion.

Definition 1.2. Let $e_1, \dots, e_n$ be the standard basis for V . For any subset $I = \{i_1 < \cdots < i_k\} \subseteq \{1, \dots, n\}$, the *Plucker coordinate* p_I of a k -dimensional subspace W is the coefficient of $e_{i_1} \wedge \cdots \wedge e_{i_k}$ in $w_1 \wedge \cdots \wedge w_k$, where $(w_1, \dots, w_k)$ is any basis for W . Moreover, they are the coefficients $p_{i_1, \dots, i_k}$ satisfying

$$w_1 \wedge \cdots \wedge w_k = \sum_{1 \leq i_1 < \cdots < i_k \leq n} p_{i_1, \dots, i_k} e_{i_1} \wedge \cdots \wedge e_{i_k}.$$

These coordinates are also defined up to a common nonzero scalar – replacing the basis amounts to scaling by the determinant of the change of basis matrix. Thus,

$$[p_{i_1, \dots, i_k}] \in \mathbb{P} \left(\bigwedge^k V \right).$$

Equivalently, p_I is the $k \times k$ minor of the matrix $(w_1 \mid \cdots \mid w_k)$ with row set I .

We now characterize the image of the Plucker embedding.

Definition 1.3. An element $\eta \in \bigwedge^k V$ is *decomposable* (or *totally decomposable*, or *simple*) if it can be written in the form $\eta = v_1 \wedge \cdots \wedge v_k$ for some $v_1, \dots, v_k \in V$. The image of the Plucker embedding is the locus of decomposable elements in $\mathbb{P}(\bigwedge^k V)$.

Remark 1.4. A nonzero element $\eta \in \bigwedge^k V$ is decomposable if and only if the multiplication map

$$\wedge \eta : V \longrightarrow \bigwedge^{k+1} V$$

has kernel of dimension at least k . This demonstrates the image of the Plucker embedding as the zero locus of the $(n - k + 1) \times (n - k + 1)$ minors of the matrix representing this map, showing that $\text{Gr}(k, V)$ is a projective variety.

In fact, the ideal of the Grassmannian is generated in degree 2:

Theorem 1.5. The image of the Plucker embedding $\text{Gr}(k, V) \hookrightarrow \mathbb{P}(\bigwedge^k V)$ is cut out by quadratic polynomials in the Plucker coordinates, called the *Plucker relations*. For any $v_1, \dots, v_k, w_1, \dots, w_k \in V$, the identity

$$(v_1 \wedge \dots \wedge v_k) \cdot (w_1 \wedge \dots \wedge w_k) - \sum_{s=1}^k (v_1 \wedge \dots \wedge v_{k-1} \wedge w_s) \cdot (w_1 \wedge \dots \wedge \widehat{w_s} \wedge \dots \wedge w_k \wedge v_k) = 0$$

holds on the image of the Plucker embedding. More precisely, these relations generate the homogeneous ideal of $\text{Gr}(k, V)$ in $\text{Sym}^*(\bigwedge^k V)$.

Proof. The quadratic relations follow from Sylvester's lemma on determinantal identities; see [Ful97, Lemma 1, §9.1]. That these generate the full ideal follows from the fact that the quadratic relations generate a prime ideal, as shown in [Ful97, §8.4], combined with the Nullstellensatz. $\square$

In coordinates, these take the following form. If $I = (i_1, \dots, i_k)$ and $J = (j_1, \dots, j_k)$ are two index sets, then

$$\sum_s \pm p_{i_1, \dots, i_{k-1}, j_s} \cdot p_{j_1, \dots, \widehat{j_s}, \dots, j_k, i_k} = 0,$$

where the signs come from reordering the indices to be increasing.

Example 1.6. Consider $\text{Gr}(2, 4)$, the Grassmannian of 2-planes in a 4-dimensional space. The Plucker embedding sends $\text{Gr}(2, 4) \hookrightarrow \mathbb{P}(\bigwedge^2 V) \cong \mathbb{P}^5$, with homogeneous coordinates $p_{12}, p_{13}, p_{14}, p_{23}, p_{24}, p_{34}$. The Grassmannian $\text{Gr}(2, 4)$ has dimension $2 \cdot 2 = 4$, so it is a codimension-1 subvariety of $\mathbb{P}^5$, i.e. a hypersurface. We have a single Plucker relation

$$p_{12} p_{34} - p_{13} p_{24} + p_{14} p_{23} = 0.$$

This can be seen by observing that a 4×4 matrix formed by repeating the two rows of a 2×4 matrix A has determinant zero, and expanding this determinant as a sum of products of 2×2 minors. Alternatively, this follows from the relation $\eta \wedge \eta = 0$ for $\eta \in \bigwedge^2 V$.

Remark 1.7. For $\text{Gr}(2, n)$ in general, the element $\eta = \sum_{a < b} p_{ab} e_a \wedge e_b \in \bigwedge^2 V$ is decomposable if and only if $\eta \wedge \eta = 0 \in \bigwedge^4 V$ (provided $\text{char}(F) \neq 2$). The coefficients of $\eta \wedge \eta$ give the Plucker relations.

We now describe how the Grassmannian is covered by affine open subsets.

Proposition 1.8. For each index set $I = \{i_1 < \dots < i_k\} \subseteq \{1, \dots, n\}$, define the open subset

$$U_I = \{\Lambda \in \text{Gr}(k, V) : p_I(\Lambda) \neq 0\}.$$

Then $U_I \cong \mathbb{A}^{k(n-k)}$, and the U_I 's cover $\text{Gr}(k, V)$.

Proof. Without loss of generality take $I = \{1, \dots, k\}$. Any $\Lambda \in U_I$ can be uniquely represented as the row space of a matrix of the form

$$B = (I_k | A)$$

where I_k is the $k \times k$ identity matrix and $A = (a_{s,t})$ is a $k \times (n - k)$ matrix. The entries of A give coordinates on U_I , establishing the isomorphism $U_I \cong \mathbb{A}^{k(n-k)}$. Since any k -plane has at least one nonvanishing $k \times k$ minor, the open sets U_I cover $\text{Gr}(k, V)$. $\square$

Remark 1.9. On the chart U_I (with $I = \{1, \dots, k\}$), the entry $a_{s,t}$ of A is, up to sign, the ratio $p_{1, \dots, \hat{s}, \dots, k, k+t} / p_{1, \dots, k}$ of Plucker coordinates ([EH16, Proposition 3.2]). In particular, the Plucker coordinates restrict to regular functions on each chart.

Corollary 1.10. The Grassmannian $\text{Gr}(k, V)$ is a smooth projective variety of dimension $k(n - k)$.

Proof. Projectivity follows from the Plucker embedding (Theorem 1.5). The affine covering of Proposition 1.8 shows $\text{Gr}(k, V)$ has dimension $k(n - k)$. Smoothness follows from the fact that $\text{GL}(V)$ acts transitively on $\text{Gr}(k, V)$ by linear automorphisms. $\square$

Remark 1.11. Corollary 1.10 will be used throughout this appendix:

- ▷ Smoothness guarantees Poincare duality, which lets us pass between Schubert varieties and Schubert classes in cohomology (Definition 1.23).
- ▷ Projectivity ensures intersection numbers are well-defined finite integers (Section 2.0.1).
- ▷ Over $F = \mathbb{C}$, smoothness and projectivity together give the identification $A^*(\text{Gr}(k, n)) \cong H^*(\text{Gr}(k, n); \mathbb{Z})$ of the Chow ring with the integral cohomology ring (Remark 2.3).

Definition 1.12. Over $\text{Gr}(k, V)$ there is a tautological exact sequence of vector bundles

$$0 \longrightarrow S \longrightarrow V \otimes \mathcal{O}_{\text{Gr}} \longrightarrow Q \longrightarrow 0,$$

where S is the *tautological (or universal) subbundle* of rank k , with fibre $S_\Lambda = \Lambda$ over a point $\Lambda \in \text{Gr}(k, V)$, and $Q = (V \otimes \mathcal{O}_{\text{Gr}})/S$ is the *universal quotient bundle* of rank $n - k$. That S is indeed a vector bundle (an algebraic subbundle of the trivial bundle) can be checked on the affine cover of Proposition 1.8; see [EH16, Proposition 3.3].

Example 1.13. When $k = 1$, so $\text{Gr}(1, n) = \mathbb{P}(V) \cong \mathbb{P}^{n-1}$, the tautological subbundle S is the line bundle $\mathcal{O}_{\mathbb{P}}(-1)$.

1.2. Schubert cells and the CW structure. Our goal is to decompose $\text{Gr}_{\mathbb{C}}(k, n)$ into cells, each of which is isomorphic to some affine space $\mathbb{C}^m$. These cells will be indexed by partitions (equivalently, Young diagrams).

Notation 1.14. A *partition* $\lambda = (\lambda_1 \geq \dots \geq \lambda_k)$ *fits inside a* $k \times (n - k)$ *box* if $\lambda_1 \leq n - k$. We write $\lambda \subseteq k \times (n - k)$ for this condition. The *size* of λ is $|\lambda| = \lambda_1 + \dots + \lambda_k$. We depict λ as a Young diagram with k rows, where the i th row has λ_i boxes, left-justified. We write $\emptyset$ for the empty partition $(0, \dots, 0)$.

Example 1.15. The partitions fitting inside a 2×2 box are:

$$\emptyset, \quad (1), \quad (1, 1), \quad (2), \quad (2, 1), \quad (2, 2).$$

These have sizes 0, 1, 2, 2, 3, 4 respectively. There are $\binom{n}{k}$ partitions fitting inside a $k \times (n - k)$ box in general.

Definition 1.16. Fix the standard complete flag

$$F_\bullet: \quad F_0 = \{0\} \subset F_1 = \langle e_1 \rangle \subset F_2 = \langle e_1, e_2 \rangle \subset \dots \subset F_n = \mathbb{C}^n,$$

where $e_1, \dots, e_n$ is the standard basis. For a partition $\lambda \subseteq k \times (n - k)$, we define the *Schubert cell* $\Omega_\lambda^\circ := \{V \in \text{Gr}_{\mathbb{C}}(k, n) : \dim(V \cap F_{n-k+i-\lambda_i}) = i \text{ and } \dim(V \cap F_{n-k+i-\lambda_i-1}) = i - 1, \quad 1 \leq i \leq k\}$. In other words, Ω_λ° consists of those k -planes V whose dimension of intersection with the flag "jumps" at the positions dictated by λ .

Remark 1.17. An equivalent formulation uses the sequence $a_i = n - k + i - \lambda_i$ for $1 \leq i \leq k$. The conditions then say that the first time $\dim(V \cap F_j) = i$ is at $j = a_i$. Some references index Schubert cells by the sequence $(a_1, \dots, a_k)$ rather than the partition λ . We follow the conventions of [Ful97] and [EH16] in using partitions.

Proposition 1.18. Each Schubert cell Ω_λ° is isomorphic to $\mathbb{C}^{k(n-k)-|\lambda|}$ as a complex manifold. Explicitly, Ω_λ° is parametrised by a reduced row echelon form with free entries precisely in the positions determined by λ .

Proof. Let $\lambda \subseteq k \times (n-k)$ and define $a_i = n - k + i - \lambda_i$ for $1 \leq i \leq k$, so that $1 \leq a_1 < a_2 < \dots < a_k \leq n$. A k -plane $V \in \Omega_\lambda^\circ$ has a unique basis $v_1, \dots, v_k$ such that, when we write the v_i as column vectors and form the $k \times n$ matrix M with v_i as the i th row:

- ▷ the a_i th column of M is the standard basis vector e_i (the pivot columns),
- ▷ for $j < a_i$ and $j \neq a_1, \dots, a_{i-1}$, the (i, j) -entry of M is a free complex parameter,
- ▷ for $j > a_i$, the (i, j) -entry is zero.

The number of free entries is exactly $\sum_{i=1}^k ((n-k) - \lambda_i) = k(n-k) - |\lambda|$, since the i th row contributes λ_i zeros to the right of the pivot (after removing later pivot columns) and $(a_i - i) = (n-k) - \lambda_i$ free entries to its left (after removing earlier pivot columns). The map sending these $k(n-k) - |\lambda|$ free parameters to the row span of the resulting matrix defines an isomorphism $\mathbb{C}^{k(n-k)-|\lambda|} \xrightarrow{\sim} \Omega_\lambda^\circ$. $\square$

Example 1.19. We describe the Schubert cells of $\text{Gr}_{\mathbb{C}}(2, 4)$ via the reduced row echelon parametrization. A point of $\text{Gr}_{\mathbb{C}}(2, 4)$ is a 2-plane in $\mathbb{C}^4$, represented by a 2×4 matrix up to left multiplication by $\text{GL}_2(\mathbb{C})$. The standard flag is $F_1 = \langle e_1 \rangle \subset F_2 = \langle e_1, e_2 \rangle \subset F_3 = \langle e_1, e_2, e_3 \rangle \subset F_4 = \mathbb{C}^4$. The six cells are as follows

λ	$k(n-k) - \lambda $	pivot positions	reduced row echelon form
$\emptyset$	4	(3, 4)	$\begin{pmatrix} * & * & 1 & 0 \\ * & * & 0 & 1 \end{pmatrix}$
(1)	3	(2, 4)	$\begin{pmatrix} * & 1 & 0 & 0 \\ * & 0 & * & 1 \end{pmatrix}$
(1, 1)	2	(2, 3)	$\begin{pmatrix} * & 1 & 0 & 0 \\ * & 0 & 1 & 0 \end{pmatrix}$
(2)	2	(1, 4)	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & * & * & 1 \end{pmatrix}$
(2, 1)	1	(1, 3)	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & * & 1 & 0 \end{pmatrix}$
(2, 2)	0	(1, 2)	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$

Here each $*$ denotes a free complex parameter. One checks that the six cells have dimensions 4, 3, 2, 2, 1, 0 respectively, giving $\binom{4}{2} = 6$ cells in total.

Definition 1.20. The *Schubert variety* Ω_λ is the closure of the Schubert cell in the Zariski topology:

$$\Omega_\lambda := \overline{\Omega_\lambda^\circ}.$$

We can have a description without reference to the closure:

$$\Omega_\lambda = \{V \in \text{Gr}_{\mathbb{C}}(k, n) : \dim(V \cap F_{n-k+i-\lambda_i}) \geq i, 1 \leq i \leq k\}.$$

This is a closed subvariety of $\text{Gr}_{\mathbb{C}}(k, n)$ of complex codimension $|\lambda|$.

Theorem 1.21. The Grassmannian admits a cell decomposition

$$\text{Gr}_{\mathbb{C}}(k, n) = \coprod_{\lambda \subseteq k \times (n-k)} \Omega_\lambda^\circ.$$

The closure relations are governed by containment of Young diagrams:

$$\Omega_\mu^\circ \subseteq \overline{\Omega_\lambda} \iff \lambda_i \leq \mu_i, \forall i \iff \lambda \subseteq \mu \text{ as Young diagrams.}$$

Proof. The cells partition $\text{Gr}_\mathbb{C}(k, n)$ because every k -plane V has a unique reduced row echelon form with respect to the standard flag. Indeed, the pivot positions $(a_1, \dots, a_k)$ are determined by the sequence of jumping dimensions $\dim(V \cap F_j)$, which gives a unique partition λ via $\lambda_i = (n - k + i) - a_i$. The constraint $\lambda_i \geq \lambda_{i+1}$ follows from $a_{i+1} > a_i$, and $\lambda_1 \leq n - k$ since $a_1 \geq 1$, so λ indeed fits inside a $k \times (n - k)$ box.

For the closure relation, observe that the Schubert variety Ω_λ consists of all k -planes satisfying $\dim(V \cap F_{n-k+i-\lambda_i}) \geq i$ (a closed condition). If $V \in \Omega_\mu^\circ \subseteq \Omega_\lambda$, then for each i we need $\dim(V \cap F_{n-k+i-\lambda_i}) \geq i$, but by definition the first time $\dim(V \cap F_j) = i$ is at $j = n - k + i - \mu_i$. So we need $n - k + i - \lambda_i \geq n - k + i - \mu_i$, i.e. $\lambda_i \leq \mu_i$. Conversely, if $\lambda \subseteq \mu$, one checks that Ω_μ° lies in the closure by exhibiting explicit one-parameter families degenerating from Ω_λ° to Ω_μ° (set free parameters to zero in the row echelon form). $\square$

Corollary 1.22. The Schubert decomposition gives $\text{Gr}_\mathbb{C}(k, n)$ the structure of a CW complex with cells only in even real dimensions. In particular,

- (1) The integral cohomology $H^*(\text{Gr}_\mathbb{C}(k, n); \mathbb{Z})$ is free abelian.
- (2) There is no odd-dimensional cohomology.
- (3) There is one Schubert class σ_λ in degree $2|\lambda|$ for each partition $\lambda \subseteq k \times (n - k)$.

Proof. Let $d = k(n - k)$. By [Proposition 1.18](#), each Schubert cell satisfies

$$\Omega_\lambda^\circ \cong \mathbb{C}^{d-|\lambda|}.$$

Thus Ω_λ° has real dimension $2(d - |\lambda|)$, which is even. The Schubert decomposition is an affine cell decomposition of $\text{Gr}_\mathbb{C}(k, n)$, and the closure relation says that if $\Omega_\mu^\circ \subseteq \overline{\Omega_\lambda}$, then $\lambda \subseteq \mu$. In particular, if Ω_μ° lies in the boundary of Ω_λ° , then $|\mu| > |\lambda|$, so

$$\dim_{\mathbb{R}} \Omega_\mu^\circ < \dim_{\mathbb{R}} \Omega_\lambda^\circ.$$

Hence the Schubert decomposition gives a finite CW decomposition of $\text{Gr}_\mathbb{C}(k, n)$ with only even-dimensional cells.

The cellular chain complex is therefore concentrated in even degrees. Since cellular boundary maps decrease degree by 1, every cellular boundary map is zero. Hence the integral homology groups are free abelian, with one generator for each Schubert cell of the corresponding real dimension, and all odd-dimensional homology vanishes. By the universal coefficient theorem, the same is true for cohomology.

Finally, the Schubert variety Ω_λ has complex codimension $|\lambda|$, so its Poincare dual

$$\sigma_\lambda = [\Omega_\lambda]^*$$

lies in degree $2|\lambda|$. These classes give one cohomology generator in degree $2|\lambda|$ for each partition $\lambda \subseteq k \times (n - k)$. $\square$

Definition 1.23. Let

$$\sigma_\lambda \in H^{2|\lambda|}(\text{Gr}_\mathbb{C}(k, n); \mathbb{Z})$$

denote the cohomology class Poincare dual to the Schubert variety Ω_λ . Explicitly, if $[\Omega_\lambda] \in H_{2(k(n-k)-|\lambda|)}(\text{Gr}_\mathbb{C}(k, n); \mathbb{Z})$ denotes the fundamental class of Ω_λ , then σ_λ is the unique class satisfying

$$\sigma_\lambda \cap [\text{Gr}_\mathbb{C}(k, n)] = [\Omega_\lambda].$$

By [Corollary 1.22](#), the set $\{\sigma_\lambda\}_{\lambda \subseteq k \times (n-k)}$ forms a $\mathbb{Z}$ -basis for $H^*(\text{Gr}_\mathbb{C}(k, n); \mathbb{Z})$.

Example 1.24. The cohomology of $\mathrm{Gr}_{\mathbb{C}}(2, 4)$ is free abelian with Betti numbers

$$b_0 = 1, \quad b_2 = 1, \quad b_4 = 2, \quad b_6 = 1, \quad b_8 = 1,$$

and all odd Betti numbers zero. In particular $\chi(\mathrm{Gr}_{\mathbb{C}}(2, 4)) = 6 = \binom{4}{2}$.

2. THE COHOMOLOGY RING AND SCHUBERT CALCULUS

Since the Schubert classes $\{\sigma_{\lambda}\}$ form an additive basis for $H^*(\mathrm{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$, the ring structure is encoded in the *structure constants* for multiplication.

Definition 2.1. The *Littlewood–Richardson coefficients* $c_{\lambda\mu}^{\nu}$ are the non-negative integers defined by

$$\sigma_{\lambda} \cdot \sigma_{\mu} = \sum_{\nu} c_{\lambda\mu}^{\nu} \sigma_{\nu}$$

in $H^*(\mathrm{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$. The sum is over all $\nu \subseteq k \times (n - k)$ with $|\nu| = |\lambda| + |\mu|$.

2.0.1. *Intersection-theoretic interpretation.* The product in cohomology can be computed geometrically via intersection of Schubert varieties defined with respect to flags in general position.

Proposition 2.2. Let λ, μ be partitions with $|\lambda| + |\mu| = k(n - k)$ (in other words their Schubert varieties have complementary dimension). Let $F_{\bullet}$ and $G_{\bullet}$ be two complete flags in $\mathbb{C}^n$ in general position. Then the Schubert varieties $\Omega_{\lambda}(F_{\bullet})$ and $\Omega_{\mu}(G_{\bullet})$ meet transversally in a finite number of points, and

$$\int_{\mathrm{Gr}_{\mathbb{C}}(k, n)} \sigma_{\lambda} \cdot \sigma_{\mu} = \#(\Omega_{\lambda}(F_{\bullet}) \cap \Omega_{\mu}(G_{\bullet})).$$

Proof. This follows from the general fact that, on a smooth projective variety, Poincaré duals of subvarieties can be multiplied in cohomology by intersecting the subvarieties after moving them into general position. The key input is that $\mathrm{GL}_n(\mathbb{C})$ acts transitively on complete flags, so we can always achieve general position, and a dimension count shows transversality in the complementary-dimension case. See [EH16, Chapter 4] for details. $\square$

Remark 2.3. Since $\mathrm{Gr}_{\mathbb{C}}(k, n)$ is a smooth complex projective variety, the cycle map induces an isomorphism between the Chow ring $A^*(\mathrm{Gr}_{\mathbb{C}}(k, n))$ and the integral cohomology ring $H^*(\mathrm{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$. All the multiplication formulas we prove below therefore hold equally in the Chow ring.

2.0.2. *Special Schubert classes.* Among all the Schubert classes, those indexed by single-row or single-column partitions play a distinguished role.

Definition 2.4. The *special Schubert classes* are:

- ▷ $\sigma_p := \sigma_{(p)}$ for $1 \leq p \leq n - k$ (single-row partitions): the class of the locus of k -planes V satisfying $\dim(V \cap F_{n-k+1-p}) \geq 1$, i.e. V meets the fixed subspace $F_{n-k+1-p}$ nontrivially.
- ▷ $\sigma_{1^p} := \sigma_{(\underbrace{1, \dots, 1}_p)}$ for $1 \leq p \leq k$ (single-column partitions).

Example 2.5. On $\mathrm{Gr}_{\mathbb{C}}(2, 4)$, the class σ_1 is Poincaré dual to the Schubert variety

$$\Omega_{(1)} = \{V \in \mathrm{Gr}_{\mathbb{C}}(2, 4) : V \cap F_2 \neq 0\},$$

the locus of 2-planes in $\mathbb{C}^4$ meeting the fixed 2-plane $F_2 = \langle e_1, e_2 \rangle$ nontrivially. This is a codimension-1 condition, and indeed $\sigma_1 \in H^2(\mathrm{Gr}_{\mathbb{C}}(2, 4); \mathbb{Z})$.

Under the Plücker embedding $\mathrm{Gr}_{\mathbb{C}}(2, 4) \hookrightarrow \mathbb{P}^5$, the class σ_1 is the hyperplane class, and $\Omega_{(1)}$ is the intersection of $\mathrm{Gr}_{\mathbb{C}}(2, 4)$ with a hyperplane in $\mathbb{P}^5$. This can be seen from the Plücker relation $p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$ and the fact that meeting F_2 is a linear condition on the Plücker coordinates.

2.0.3. *The Pieri rule.* The Pieri rule gives an explicit formula for the product of any Schubert class with a special Schubert class. Together with the Giambelli formula below, it determines the entire ring structure.

Theorem 2.6 (Pieri rule). For any partition $\lambda \subseteq k \times (n - k)$ and any $1 \leq p \leq n - k$,

$$\sigma_p \cdot \sigma_\lambda = \sum_{\mu} \sigma_\mu,$$

where the sum is over all partitions $\mu \subseteq k \times (n - k)$ satisfying:

- (1) $|\mu| = |\lambda| + p$,
- (2) $\mu_i \geq \lambda_i$ for all i (i.e. μ is obtained by adding boxes to λ),
- (3) $\mu_1 \leq n - k$ and $\mu_i \leq \lambda_{i-1}$ for $i \geq 2$ (no two added boxes lie in the same column).

Equivalently, μ/λ is a *horizontal strip* of size p : a skew shape with at most one box in each column.

Proof. See [Ful97, §9.4] or [EH16, Chapter 4]. I do not really want to write this down. $\square$

Theorem 2.7 (Dual Pieri rule). For any partition $\lambda \subseteq k \times (n - k)$ and any $1 \leq p \leq k$,

$$\sigma_{1^p} \cdot \sigma_\lambda = \sum_{\mu} \sigma_\mu,$$

where the sum is over all $\mu \subseteq k \times (n - k)$ with $|\mu| = |\lambda| + p$ such that μ/λ is a *vertical strip* of size p : a skew shape with at most one box in each row.

Proof. This follows from the Pieri rule by duality. The involution $\lambda \mapsto \lambda'$ (transposing the Young diagram) exchanges $\text{Gr}_{\mathbb{C}}(k, n)$ with $\text{Gr}_{\mathbb{C}}(n - k, n)$ and interchanges σ_p with σ_{1^p} . Alternatively, see [Ful97, §9.4]. $\square$

Example 2.8. We show the Pieri rule on $\text{Gr}_{\mathbb{C}}(2, 4)$, where partitions fit inside a 2×2 box. The special Schubert classes are σ_1 and σ_2 . We compute all products $\sigma_1 \cdot \sigma_\lambda$ by finding partitions $\mu \subseteq 2 \times 2$ with $|\mu| = |\lambda| + 1$ such that μ/λ is a horizontal strip (at most one added box per column):

$$\begin{aligned} \sigma_1 \cdot \sigma_{\emptyset} &= \sigma_1, \\ \sigma_1 \cdot \sigma_1 &= \sigma_2 + \sigma_{1,1}, \\ \sigma_1 \cdot \sigma_{1,1} &= \sigma_{2,1}, \\ \sigma_1 \cdot \sigma_2 &= \sigma_{2,1}, \\ \sigma_1 \cdot \sigma_{2,1} &= \sigma_{2,2}, \\ \sigma_1 \cdot \sigma_{2,2} &= 0. \end{aligned}$$

For instance, $\sigma_1 \cdot \sigma_1 = \sigma_2 + \sigma_{1,1}$ because one can add a single box to (1) either at the end of the first row to obtain (2), or as a new second row to obtain (1, 1); both are horizontal strips. On the other hand, $\sigma_1 \cdot \sigma_{2,2} = 0$ because no box can be added to the partition (2, 2) while remaining inside the 2×2 box.

The products with σ_2 are computed similarly; for instance $\sigma_2 \cdot \sigma_{1,1} = 0$ because the only partition of size 4 in the 2×2 box is (2, 2), but (2, 2)/(1, 1) has two boxes in its second column, violating the horizontal strip condition.

To double check, observe that $\sigma_1^2 = \sigma_2 + \sigma_{1,1}$ and $\sigma_1^3 = \sigma_1(\sigma_2 + \sigma_{1,1}) = \sigma_{2,1} + \sigma_{2,1} = 2\sigma_{2,1}$, so

$$\int_{\text{Gr}_{\mathbb{C}}(2,4)} \sigma_1^4 = \int_{\text{Gr}_{\mathbb{C}}(2,4)} 2\sigma_{2,1} \cdot \sigma_1 = 2.$$

This is the number of lines in $\mathbb{P}^3$ meeting four lines in general position.

2.0.4. *The Giambelli formula.* The Giambelli formula expresses an arbitrary Schubert class as a determinant in the special Schubert classes. Combined with the Pieri rule, it gives an explicit algorithm for multiplying any two Schubert classes.

Theorem 2.9 (Giambelli formula). For any partition $\lambda = (\lambda_1, \dots, \lambda_k) \subseteq k \times (n - k)$,

$$\sigma_\lambda = \det(\sigma_{\lambda_i - i + j})_{1 \leq i, j \leq k},$$

where $\sigma_0 = 1$ and $\sigma_m = 0$ for $m < 0$ or $m > n - k$

Proof. This can be proved by induction on the number of parts of λ , using the Pieri rule and the Jacobi–Trudi identity from symmetric function theory. See [Ful97, Chapter 10]. $\square$

Example 2.10. We apply the Giambelli formula on $\text{Gr}_{\mathbb{C}}(2, 4)$. Here $k = 2$, so the formula gives each Schubert class as a 2×2 determinant in the special classes σ_1, σ_2 (with $\sigma_0 = 1$ and $\sigma_m = 0$ for $m < 0$ or $m > 2$)

$$\sigma_{(\lambda_1, \lambda_2)} = \det \begin{pmatrix} \sigma_{\lambda_1} & \sigma_{\lambda_1+1} \\ \sigma_{\lambda_2-1} & \sigma_{\lambda_2} \end{pmatrix}.$$

For the single-row partitions $\lambda = (1)$ and $\lambda = (2)$, the formula gives

$$\sigma_1 = \det \begin{pmatrix} \sigma_1 & \sigma_2 \\ \sigma_{-1} & \sigma_0 \end{pmatrix} = \sigma_1 \cdot 1 - \sigma_2 \cdot 0 = \sigma_1,$$

$$\sigma_2 = \det \begin{pmatrix} \sigma_2 & \sigma_3 \\ \sigma_{-1} & \sigma_0 \end{pmatrix} = \sigma_2 \cdot 1 - 0 \cdot 0 = \sigma_2.$$

The non-trivial cases express the remaining Schubert classes as polynomials in σ_1 and σ_2 ,

$$\sigma_{1,1} = \det \begin{pmatrix} \sigma_1 & \sigma_2 \\ \sigma_0 & \sigma_1 \end{pmatrix} = \sigma_1^2 - \sigma_2,$$

$$\sigma_{2,1} = \det \begin{pmatrix} \sigma_2 & \sigma_3 \\ \sigma_0 & \sigma_1 \end{pmatrix} = \sigma_1 \sigma_2,$$

$$\sigma_{2,2} = \det \begin{pmatrix} \sigma_2 & \sigma_3 \\ \sigma_1 & \sigma_2 \end{pmatrix} = \sigma_2^2.$$

Corollary 2.11. The classes $\sigma_1, \sigma_2, \dots, \sigma_{n-k}$ generate $H^*(\text{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$ as a ring.

Proof. The Giambelli formula writes every σ_λ as a polynomial in $\sigma_1, \dots, \sigma_{n-k}$, and these classes form an additive basis. $\square$

2.0.5. *The Littlewood–Richardson rule.* The Pieri rule handles products with special Schubert classes. For the general product, we need the Littlewood–Richardson rule.

Definition 2.12. Let ν/λ be a skew shape and μ a partition with $|\mu| = |\nu| - |\lambda|$. A *Littlewood–Richardson tableau* of shape ν/λ and content μ is a filling of the boxes of ν/λ with μ_1 copies of 1, μ_2 copies of 2, and so on, such that:

- (1) the entries weakly increase along each row (left to right),
- (2) the entries strictly increase down each column,
- (3) the *reverse reading word* (reading right to left across rows, top to bottom) is a *lattice word*.
For every initial segment of the word and every i , the number of i 's is at least the number of $(i + 1)$'s.

Theorem 2.13 (Littlewood–Richardson rule). For any two partitions λ, μ with $\lambda, \mu, \nu \subseteq k \times (n - k)$, the Littlewood–Richardson coefficient $c_{\lambda\mu}^\nu$ equals the number of Littlewood–Richardson (LR) tableaux of shape ν/λ and content μ .

Proof. We will not prove this here; see [Ful97, Chapter 5 and §9.4]. $\square$

Example 2.14. As a sample computation, we can verify using the Littlewood–Richardson rule that

$$\sigma_{2,1}^2 = \sigma_{3,3} + 2\sigma_{3,2,1} + \sigma_{2,2,2} \quad \text{in } H^*(\text{Gr}_{\mathbb{C}}(3, 6); \mathbb{Z}).$$

The coefficient 2 for $\sigma_{3,2,1}$ arises because the skew shape $(3, 2, 1)/(2, 1)$ admits two distinct LR tableaux of content $(2, 1)$. This is a non-Pieri product — neither factor is a single-row or single-column partition.

Example 2.15. We give the full multiplication table for $H^*(\text{Gr}_{\mathbb{C}}(2, 4); \mathbb{Z})$. The six Schubert classes are $1 = \sigma_{\emptyset}$, σ_1 , $\sigma_{1,1}$, σ_2 , $\sigma_{2,1}$, $\sigma_{2,2}$, in cohomology degrees 0, 2, 4, 4, 6, 8 respectively. Since products of classes whose degrees exceed $2k(n - k) = 8$ vanish, many entries are zero for degree reasons alone. The non-trivial products are:

$\cdot$	1	σ_1	σ_2	$\sigma_{1,1}$	$\sigma_{2,1}$	$\sigma_{2,2}$
1	1	σ_1	σ_2	$\sigma_{1,1}$	$\sigma_{2,1}$	$\sigma_{2,2}$
σ_1		$\sigma_2 + \sigma_{1,1}$	$\sigma_{2,1}$	$\sigma_{2,1}$	$\sigma_{2,2}$	0
σ_2			$\sigma_{2,2}$	0	0	0
$\sigma_{1,1}$				$\sigma_{2,2}$	0	0
$\sigma_{2,1}$					0	0
$\sigma_{2,2}$						0

The products with σ_1 and σ_2 follow from the Pieri rule (Example 2.8). The product $\sigma_{1,1}^2 = \sigma_{2,2}$ follows from the dual Pieri rule $\sigma_{1,1} = \sigma_{1^2}$, and the only partition $\mu \subseteq 2 \times 2$ with $|\mu| = 4$ such that $\mu/(1, 1)$ is a vertical strip is $\mu = (2, 2)$. All remaining products vanish because their degrees exceed $8 = 2k(n - k)$.

The vanishing $\sigma_2 \cdot \sigma_{1,1} = 0$ is notable, as σ_2 represents the locus of lines passing through a fixed point p , while $\sigma_{1,1}$ represents the locus of lines contained in a fixed plane H . For general choices, $p \notin H$, so there is no line through p contained in H .

The class $\sigma_{2,2}$ is the class of a point (the unique 2-plane spanned by two given basis vectors), and $\int_{\text{Gr}_{\mathbb{C}}(2,4)} \sigma_{2,2} = 1$. The intersection pairing between complementary degrees therefore gives

$$\int \sigma_1 \cdot \sigma_{2,1} = 1, \quad \int \sigma_2^2 = 1, \quad \int \sigma_{1,1}^2 = 1, \quad \int \sigma_2 \cdot \sigma_{1,1} = 0.$$

Using the Giambelli relations $\sigma_{1,1} = \sigma_1^2 - \sigma_2$ and $\sigma_{2,1} = \sigma_1\sigma_2$ from Example 2.10, the entire ring is generated by σ_1 and σ_2 ,

$$H^*(\text{Gr}_{\mathbb{C}}(2, 4); \mathbb{Z}) \cong \mathbb{Z}[\sigma_1, \sigma_2] / (\sigma_1^3 - 2\sigma_1\sigma_2, \sigma_1^2\sigma_2 - \sigma_2^2),$$

where $\deg \sigma_1 = 2$ and $\deg \sigma_2 = 4$. The first relation comes from $\sigma_1 \cdot \sigma_{1,1} = \sigma_1 \cdot \sigma_2$, i.e. $\sigma_1^3 - \sigma_1\sigma_2 = \sigma_1\sigma_2$, while the second is $\sigma_2 \cdot \sigma_{1,1} = 0$, i.e. $\sigma_1^2\sigma_2 = \sigma_2^2$.

Remark 2.16. Under the identification $\text{Gr}_{\mathbb{C}}(2, 4) \cong \mathbb{G}(1, 3)$ (the Grassmannian of lines in $\mathbb{P}^3$), the Schubert class σ_1 represents the locus of lines meeting a fixed line, and σ_2 represents the locus of lines passing through a fixed point. The products above encode enumerative facts about lines in $\mathbb{P}^3$:

- ▷ $\int \sigma_1^4 = \int (\sigma_2 + \sigma_{1,1})^2 = \int (\sigma_2^2 + \sigma_{1,1}^2) = 2$: there are 2 lines in $\mathbb{P}^3$ meeting four lines in general position.
- ▷ $\int \sigma_1^2 \cdot \sigma_2 = \int \sigma_{2,2} = 1$: there is 1 line through a given point meeting two given lines in general position.
- ▷ $\int \sigma_2^2 = 1$: there is 1 line through two given points.

3. CHERN CLASSES AS SCHUBERT CLASSES

We now connect the Schubert calculus to the theory of characteristic classes developed in our class. We show that Chern classes of the tautological bundles on $\mathrm{Gr}_{\mathbb{C}}(k, n)$ are special Schubert classes, and that the Whitney sum formula encodes all the relations in the cohomology ring.

3.0.1. The tautological and quotient bundles.

Remark 3.1. The bundle S is the same as the tautological bundle $E_k(\mathbb{C}^n) \rightarrow \mathrm{Gr}_{\mathbb{C}}(k, n)$ from Proposition 7.7. On $\mathbb{CP}^n = \mathrm{Gr}_{\mathbb{C}}(1, n+1)$, the bundle S restricts to $\mathcal{O}(-1)$ (Remark 7.10), so S^* restricts to $\mathcal{O}(1)$, whose first Chern class is the positive generator $h \in H^2(\mathbb{CP}^n; \mathbb{Z})$.

3.0.2. Chern classes in terms of Schubert classes.

Theorem 3.2. The Chern classes of S^* and Q are identified with special Schubert classes:

$$\begin{aligned} c_i(S^*) &= \sigma_{1^i} \quad \text{for } 1 \leq i \leq k, \\ c_j(Q) &= \sigma_j \quad \text{for } 1 \leq j \leq n - k. \end{aligned}$$

Proof. We prove $c_j(Q) = \sigma_j$ via a degeneracy locus argument, and the formula for S^* will then follow from the Whitney sum relation.

Any vector $v \in \mathbb{C}^n$ determines a global section s_v of the quotient bundle Q , since at a point $V \in \mathrm{Gr}_{\mathbb{C}}(k, n)$, the section is the image of v under the projection $\mathbb{C}^n \twoheadrightarrow \mathbb{C}^n/V = Q_V$. This section vanishes at V if and only if $v \in V$.

Let $r = n - k$ denote the rank of Q . Choose $r - j + 1 = n - k - j + 1$ generic vectors $v_1, \dots, v_{r-j+1} \in \mathbb{C}^n$, and let $U = \mathrm{span}(v_1, \dots, v_{r-j+1})$. These vectors induce $r - j + 1$ global sections of Q . They fail to be linearly independent at $V \in \mathrm{Gr}_{\mathbb{C}}(k, n)$ precisely when their images in $Q_V = \mathbb{C}^n/V$ are linearly dependent, equivalently when

$$V \cap U \neq 0.$$

Choosing $U = F_{n-k+1-j}$, a flag element of dimension $n - k + 1 - j$ in a complete flag, this degeneracy locus becomes

$$\{V \in \mathrm{Gr}_{\mathbb{C}}(k, n) : V \cap F_{n-k+1-j} \neq 0\} = \Omega_{(j)}.$$

Since the degeneracy locus of $r - j + 1$ generic sections of a rank- r bundle has expected codimension j and represents c_j (see [Ful97, §14.7] or [EH16, Theorem 5.3]), and a transversality argument – using the transitivity of the $\mathrm{GL}_n(\mathbb{C})$ action – shows this expected codimension is achieved, we obtain $c_j(Q) = \sigma_j$.

For S^* : the Whitney sum formula applied to the tautological sequence gives $c(S) \cdot c(Q) = 1$ (Corollary 3.3). Since $c_i(S^*) = (-1)^i c_i(S)$ (Theorem 4.4.3 I of [Hir95]), we can solve for $c_i(S^*)$ from the relation $c(S) = c(Q)^{-1}$:

$$\begin{aligned} c_1(S) &= -\sigma_1, & c_1(S^*) &= \sigma_1 = \sigma_{1^1}, \\ c_2(S) &= \sigma_1^2 - \sigma_2 = \sigma_{1,1}, & c_2(S^*) &= \sigma_{1,1} = \sigma_{1^2}, \end{aligned}$$

and in general $c_i(S^*) = \sigma_{1^i}$. This can be verified either inductively from the relation $c(S) = c(Q)^{-1}$, or independently by a degeneracy locus argument for S^* : a linear functional $\ell \in (\mathbb{C}^n)^*$ restricts to a section of S^* via $\ell|_V \in V^* = S_V^*$ at each point $V \in \mathrm{Gr}_{\mathbb{C}}(k, n)$. Choosing $k - i + 1$ generic linear functionals, their restrictions to V become linearly dependent precisely when V satisfies i independent incidence conditions with respect to the corresponding hyperplanes. The resulting degeneracy locus is the Schubert variety Ω_{1^i} , giving $c_i(S^*) = \sigma_{1^i}$ directly. $\square$

Corollary 3.3. The Whitney sum formula applied to the tautological sequence gives

$$c(S) \cdot c(Q) = 1$$

in $H^*(\mathrm{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$.

Proof. By Axiom 3 of definition of chern classes, the short exact sequence $0 \rightarrow S \rightarrow \underline{\mathbb{C}}^n \rightarrow Q \rightarrow 0$ gives $c(S) \cdot c(Q) = c(\underline{\mathbb{C}}^n)$. Since $\underline{\mathbb{C}}^n$ is a trivial bundle, all its Chern classes in positive degree vanish, so $c(\underline{\mathbb{C}}^n) = 1$. $\square$

Example 3.4. We show [Theorem 3.2](#) explicitly for $\text{Gr}_{\mathbb{C}}(2, 4)$, where S has rank 2 and Q has rank 2. The Chern classes of the dual tautological bundle and the quotient bundle are

$$\begin{aligned} c_1(S^*) &= \sigma_{1^1} = \sigma_1, & c_1(Q) &= \sigma_1, \\ c_2(S^*) &= \sigma_{1^2} = \sigma_{1,1}, & c_2(Q) &= \sigma_2, \end{aligned}$$

and correspondingly $c_1(S) = -\sigma_1$ and $c_2(S) = \sigma_{1,1}$, using $c_i(S) = (-1)^i c_i(S^*)$.

The relation $c(S) \cdot c(Q) = 1$ of [Corollary 3.3](#) gives us

$$(1 - \sigma_1 + \sigma_{1,1})(1 + \sigma_1 + \sigma_2) = 1.$$

We check this degree by degree using the multiplication table of [Example 2.15](#):

$$\begin{aligned} \text{degree 2:} \quad & -\sigma_1 + \sigma_1 = 0, \\ \text{degree 4:} \quad & \sigma_{1,1} - \sigma_1^2 + \sigma_2 = \sigma_{1,1} - (\sigma_2 + \sigma_{1,1}) + \sigma_2 = 0, \\ \text{degree 6:} \quad & -\sigma_1\sigma_2 + \sigma_{1,1} \cdot \sigma_1 = -\sigma_{2,1} + \sigma_{2,1} = 0, \\ \text{degree 8:} \quad & \sigma_{1,1} \cdot \sigma_2 = 0. \end{aligned}$$

The degree-6 and degree-8 relations are consequences of the degree-4 relation together with the Pieri rule. Thus the Whitney sum formula encodes all the relations in $H^*(\text{Gr}_{\mathbb{C}}(2, 4); \mathbb{Z})$.

Setting $c_i = c_i(S^*)$ and $\bar{c}_j = c_j(Q)$, the Whitney sum formula determines each $\bar{c}_j$ in terms of the c_i . From $c(S) \cdot c(Q) = 1$, i.e. $c(Q) = c(S)^{-1}$, we solve:

$$\begin{aligned} \bar{c}_1 &= c_1 = \sigma_1, \\ \bar{c}_2 &= c_1^2 - c_2 = \sigma_1^2 - \sigma_{1,1} = \sigma_2. \end{aligned}$$

Eliminating $\bar{c}_1$ and $\bar{c}_2$ recovers the equality found in [Example 2.15](#)

$$H^*(\text{Gr}_{\mathbb{C}}(2, 4); \mathbb{Z}) \cong \mathbb{Z}[\sigma_1, \sigma_2] / (\sigma_1^3 - 2\sigma_1\sigma_2, \sigma_1^2\sigma_2 - \sigma_2^2),$$

where we identify $\sigma_1 = c_1(S^*) = c_1(Q)$ and $\sigma_2 = c_2(Q)$, and the two relations arise from the vanishing of the degree-6 and degree-8 components of $c(S) \cdot c(Q) - 1$.

3.0.3. Connection to the general theory of characteristic classes. We now explain how the results of this appendix make concrete the abstract Chern class theory from §8 of the main notes.

Proposition 3.5. For an arbitrary complex vector bundle $E \rightarrow X$ of rank k over a paracompact space X , the classifying map $f: X \rightarrow \text{Gr}_{\mathbb{C}}(k, \infty)$ (which exists by Theorem 7.11) satisfies

$$c_i(E^*) = f^* \sigma_{1^i}$$

for each $1 \leq i \leq k$. Equivalently, $c_i(E) = (-1)^i f^* \sigma_{1^i}$.

Proof. By Theorem 7.11, the bundle E is isomorphic to f^*S for some (homotopy class of) map $f: X \rightarrow \text{Gr}_{\mathbb{C}}(k, \infty)$, where S is the tautological subbundle. Therefore $E^* \cong f^*S^*$. By naturality of Chern classes (Axiom 2 of the definition in the class notes):

$$c_i(E^*) = c_i(f^*S^*) = f^*c_i(S^*).$$

Now $c_i(S^*) = \sigma_{1^i}$ by [Theorem 3.2](#), which holds on each finite Grassmannian $\text{Gr}_{\mathbb{C}}(k, N)$ and is compatible with the inclusions $\text{Gr}_{\mathbb{C}}(k, N) \hookrightarrow \text{Gr}_{\mathbb{C}}(k, N+1)$, so it passes to the colimit $\text{Gr}_{\mathbb{C}}(k, \infty)$. $\square$

Remark 3.6. Proposition 3.5 gives a more tangible geometric version of Borel's theorem (Theorem 1.8.5 in the class notes)

$$H^*(BU(k); \mathbb{Z}) = H^*(\text{Gr}_{\mathbb{C}}(k, \infty); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_k],$$

where each c_i is the special Schubert class σ_{1^i} . The Schubert calculus tells us what these generators are geometrically, since c_i is Poincaré dual to Ω_{1^i} , the locus of k -planes satisfying i independent incidence conditions with respect to a flag.

A complex characteristic class for rank k vector bundles is a cohomology class of $\text{Gr}_{\mathbb{C}}(k, \infty)$. By the Giambelli formula (Theorem 2.9), every such class is a polynomial in $\sigma_{1^1}, \dots, \sigma_{1^k}$, so every characteristic class is a polynomial in the Chern classes (basically Borel's theorem, now proved geometrically through Schubert calculus).

3.0.4. Presentation of the cohomology ring.

Theorem 3.7. The integral cohomology ring of the Grassmannian admits the presentation

$$H^*(\text{Gr}_{\mathbb{C}}(k, n); \mathbb{Z}) \cong \frac{\mathbb{Z}[c_1, \dots, c_k, \bar{c}_1, \dots, \bar{c}_{n-k}]}{(1 - c_1 + c_2 - \dots + (-1)^k c_k)(1 + \bar{c}_1 + \dots + \bar{c}_{n-k}) = 1},$$

where $c_i = c_i(S^*) = \sigma_{1^i}$ and $\bar{c}_j = c_j(Q) = \sigma_j$. The relation is shorthand for n polynomial equations obtained by expanding the product and setting each positive-degree component to zero.

Proof. We must show that the Whitney sum relations generate all relations among the c_i and $\bar{c}_j$. Surjectivity of the natural map from the polynomial ring is immediate: $c_i = c_i(S^*) = \sigma_{1^i}$ and $\bar{c}_j = c_j(Q) = \sigma_j$ are Schubert classes, and every Schubert class is a polynomial in the σ_j by the Giambelli formula (Corollary 2.11).

Let $c_i = c_i(S^*)$ and $\bar{c}_j = c_j(Q)$. Since $c_i(S) = (-1)^i c_i(S^*)$, the Whitney sum relation $c(S) \cdot c(Q) = 1$ takes the form

$$(1 - c_1 + c_2 - \dots + (-1)^k c_k)(1 + \bar{c}_1 + \dots + \bar{c}_{n-k}) = 1.$$

The first $n - k$ homogeneous components recursively express $\bar{c}_1, \dots, \bar{c}_{n-k}$ as polynomials in $c_1, \dots, c_k$. Equivalently, if h_r is defined by the generating function

$$\sum_{r \geq 0} h_r t^r = \frac{1}{1 - c_1 t + c_2 t^2 - \dots + (-1)^k c_k t^k},$$

then $\bar{c}_r = h_r$ for $1 \leq r \leq n - k$. Since Q has rank $n - k$, the remaining components give

$$h_{n-k+1} = h_{n-k+2} = \dots = h_n = 0.$$

After eliminating $\bar{c}_1, \dots, \bar{c}_{n-k}$, the quotient ring is therefore

$$R \cong \mathbb{Z}[c_1, \dots, c_k] / (h_{n-k+1}, \dots, h_n).$$

It suffices to show that R is free of rank $\binom{n}{k}$ as a $\mathbb{Z}$ -module, since we know $H^*(\text{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$ is free of this rank (Corollary 1.22). By the theory of symmetric functions (see [Ful97, §9.4]), the Schur polynomials $s_{\lambda}(c_1, \dots, c_k)$ for partitions $\lambda \subseteq k \times (n - k)$ form a free $\mathbb{Z}$ -basis of this quotient. Under the Giambelli formula (Theorem 2.9), these map to the Schubert classes σ_{λ} . Since the number of such partitions is $\binom{n}{k}$, the ring R is free of rank $\binom{n}{k}$, and the quotient map $R \rightarrow H^*(\text{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$ is an isomorphism. $\square$

Remark 3.8. One can present the cohomology ring using only one set of generators. Using the special Schubert classes $\sigma_1, \dots, \sigma_{n-k}$ (the Chern classes of Q), the Whitney sum relation determines $c(S) = c(Q)^{-1}$. Define $D_i = c_i(S) = (-1)^i \sigma_{1^i}$ by the identity

$$(1 + D_1 + D_2 + \dots)(1 + \sigma_1 + \dots + \sigma_{n-k}) = 1,$$

so that D_i is computed recursively from $D_i = -\sigma_1 D_{i-1} - \sigma_2 D_{i-2} - \cdots - \sigma_i$ (with $D_0 = 1$). Since S has rank k , the condition $D_i = 0$ for $i > k$ yields the relations

$$H^*(\mathrm{Gr}_{\mathbb{C}}(k, n); \mathbb{Z}) \cong \mathbb{Z}[\sigma_1, \dots, \sigma_{n-k}] / (D_{k+1}, D_{k+2}, \dots, D_n).$$

For instance, $D_1 = -\sigma_1$, $D_2 = \sigma_1^2 - \sigma_2$, and $D_3 = -\sigma_1^3 + 2\sigma_1\sigma_2 - \sigma_3$.

Dually, using $c_1, \dots, c_k$ (the Chern classes of S^*) as generators, the relation $c(Q) = c(S)^{-1}$ expresses each $\bar{c}_r = c_r(Q)$ as a polynomial h_r in $c_1, \dots, c_k$ via the generating function

$$\sum_{r \geq 0} h_r t^r = \frac{1}{1 - c_1 t + c_2 t^2 - \cdots + (-1)^k c_k t^k}.$$

Since Q has rank $n - k$, the conditions $h_r = 0$ for $r > n - k$ give the presentation

$$H^*(\mathrm{Gr}_{\mathbb{C}}(k, n); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_k] / (h_{n-k+1}, \dots, h_n).$$

Remark 3.9. The presentation of [Theorem 3.7](#) clarifies the transition to the infinite Grassmannian $\mathrm{Gr}_{\mathbb{C}}(k, \infty) = BU(k)$. In the dual presentation of [Remark 3.8](#), the ring $H^*(\mathrm{Gr}_{\mathbb{C}}(k, n); \mathbb{Z})$ has k generators $c_1, \dots, c_k$ and the relations $h_{n-k+1} = \cdots = h_n = 0$. As $n \rightarrow \infty$, these relations are pushed to arbitrarily high degree, with $h_r = 0$ only for $r > n - k$, and $n - k \rightarrow \infty$. In the colimit, no relations survive, and we obtain

$$H^*(BU(k); \mathbb{Z}) = H^*(\mathrm{Gr}_{\mathbb{C}}(k, \infty); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_k],$$

a polynomial ring with no relations, where $c_i = c_i(S^*) = \sigma_{1^i}$. This is Borel's theorem now obtained as a direct consequence of the finite-dimensional Schubert calculus. Note that the presentation using σ_j generators does not pass to the limit as cleanly, as both the number of generators and the number of relations tend to infinity, whereas the c_i presentation has a fixed set of k generators with relations that simply disappear.

Example 3.10. We recover the cohomology ring of $\mathbb{CP}^n$ from the presentation of [Theorem 3.7](#). Since $\mathbb{CP}^n = \mathrm{Gr}_{\mathbb{C}}(1, N)$ with $N = n + 1$, we have $k = 1$ and $N - k = n$. The tautological subbundle $S = \mathcal{O}(-1)$ is a line bundle, so there is a single Chern class $c_1 = c_1(S^*)$, and by [Remark 3.1](#) this is the positive generator $h \in H^2(\mathbb{CP}^n; \mathbb{Z})$. The quotient bundle Q has rank n , with $\bar{c}_j = c_j(Q) = \sigma_j$ for $1 \leq j \leq n$.

The Whitney sum relation $c(S) \cdot c(Q) = 1$ reads

$$(1 - h)(1 + \bar{c}_1 + \bar{c}_2 + \cdots + \bar{c}_n) = 1,$$

using $c_1(S) = -c_1(S^*) = -h$. Solving degree by degree gives $\bar{c}_j = h^j$ for each $1 \leq j \leq n$, together with the truncation condition $h^{n+1} = 0$ (from degree $n + 1$). Eliminating the $\bar{c}_j$, the presentation of [Theorem 3.7](#) reduces to

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[h] / (h^{n+1}).$$

Here $h = c_1(\mathcal{O}(1)) = \sigma_1$ is both the hyperplane class and the unique special Schubert class.

Example 3.11. We derive the ring presentation of [Theorem 3.7](#) for $\mathrm{Gr}_{\mathbb{C}}(2, 4)$ from scratch. Here $k = 2$ and $n - k = 2$, with generators $c_1, c_2, \bar{c}_1, \bar{c}_2$ as identified in [Example 3.4](#). The Whitney sum relation $c(S) \cdot c(Q) = 1$ reads

$$(1 - c_1 + c_2)(1 + \bar{c}_1 + \bar{c}_2) = 1.$$

Expanding degree by degree as in [Example 3.4](#), the degrees 2 and 4 components give $\bar{c}_1 = c_1$ and $\bar{c}_2 = c_1^2 - c_2$, eliminating the Chern classes of Q . The degrees 6 and 8 components then yield the two defining relations

$$H^*(\mathrm{Gr}_{\mathbb{C}}(2, 4); \mathbb{Z}) \cong \mathbb{Z}[c_1, c_2] / (c_1^3 - 2c_1c_2, c_1^2c_2 - c_2^2),$$

with $\deg c_1 = 2$ and $\deg c_2 = 4$. This corresponds to the presentation in terms of Schubert classes from [Example 2.15](#) under the identification $c_1 = \sigma_1$ and $c_2 = \sigma_{1,1}$, with $\sigma_2 = \bar{c}_2 = c_1^2 - c_2$.

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